# Ch-4 Cache Memory

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Performance

- Access time
- Cycle time
- Transfer rate

Physical Type

- Semi-conductor
- Magnetic
- Optical
- Magneto-optical

Physical Characteristics

2. Capacity:

Number of Words: Memory

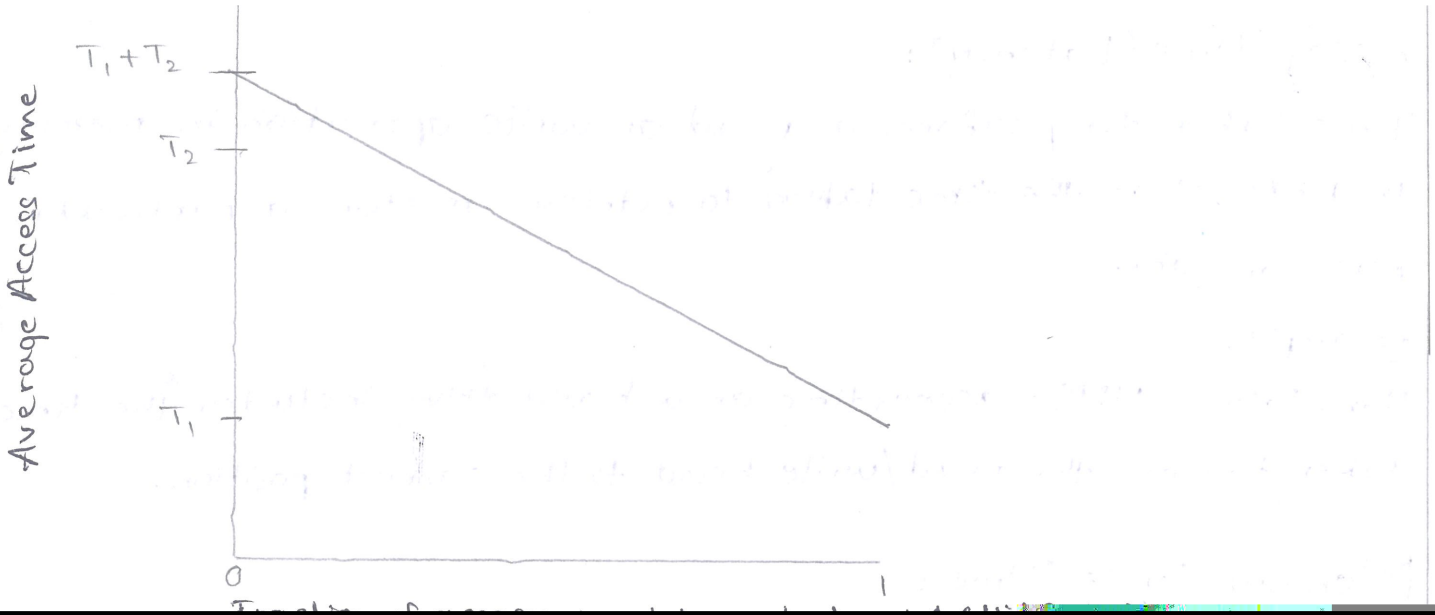
5. Performance

Access Time (Latency):

Time taken to perform a read or write operation in memory. In RAM, it is the time taken to retrieve or store a particular peice of data.

Example:

[redacted]


Average Access Time
$T_1 + T_2$
$T_2$
$T_1$
0
1

Ex: ROM (Read-Only Memory)

Memory:

- High speed memory initialized values in GPU

3. Offline Storage:

Magnetic Tape:

- Used for backup, archiving and long-term storage.

- This hierarchy allows a computer system to utilize various types of memory to achieve an optimal balance, ensuring

This image illustrates different cache and memory architecture:

[redacted]

l3 cache :- largest than L2 but slowest.it serves as backup if the is not found in l2

cache before accessing main memory